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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Sports Tournament Team Registration and Match Result System

Submitted by

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Semester: **II**

DECLARATION

I hereby declare that the project entitled **Sports Tournament Team Registration and Match Result System** submitted in partial fulfillment of the requirements for the course **BSC COMPUTER SCIENCE** is a **record of my original work** carried out under the guidance of **DR.BOOPALAN SUNDARAMOORTHY**, Technical Head , Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____

Nithish P

CERTIFICATE

This is to certify that the project entitled **Sports Tournament Team Registration and Match Result System** is a **bona fide work** carried out by NITHISH P**(I BSC CS 2025)** , **BSC COMPUTER SCIENCE**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **JAVA PROGRAMMING**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: COIMBATORE

Date:April 2026

Submitted for the Viva-Voce Examination held on _____

Guide:
(Technical Head, KGCAS)

_____

Head of the Department
(Technical Head, KGCAS)

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ABSTRACT

The Sports Tournament Team Registration and Match Result System is a Java-based application designed to manage team registrations and track match results efficiently. In many sports events, organizing teams and recording match results manually can lead to confusion, errors, and poor data management.

This system provides a simple and effective solution where users can register teams, store team details, and record match results. It ensures that all tournament information is organized in one place and can be easily accessed.

The project is developed using JavaFX for the graphical user interface and Core Java concepts such as classes, objects, and event handling. The main objective is to simplify tournament management and reduce manual effort.

1. INTRODUCTION

Managing sports tournaments manually can be time-consuming and error-prone. Organizers often face difficulties in registering teams and maintaining match results properly.

This system helps:

- Register teams easily
- Store match results
- Manage tournament data efficiently

It provides a simple GUI-based solution for handling tournament operations.

1.1 Objective

- To develop a JavaFX-based tournament system
- To register teams
- To store match results
- To reduce manual errors
- To improve organization

1.2 Overview

The system allows users to:

- Add teams
- View registered teams
- Enter match results
- View match history

1.3 Features

- Team Registration
- Match Result Recording
- Simple GUI Interface
- Easy Data Management

2. PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and the requirements of the system are identified. In the **Sports Tournament Team Registration and Match Result System**, this stage helps in understanding how tournaments are conducted, how teams are registered, and how match results are recorded, along with the features required to manage these activities efficiently.

The analysis phase focuses on identifying the issues in the existing manual system such as difficulty in managing team registrations, confusion in recording match results, and lack of proper data organization. Based on this, a computerized solution is proposed that allows users to register teams, store match details, and display results accurately, thereby improving efficiency and reducing errors.

2.1 Problem Definition

Manual tournament handling leads to:

- Loss of data
- Confusion in match results
- Difficulty in tracking teams

2.2 Project Overview

The system allows:

- Adding team names
- Recording match results
- Displaying tournament data

2.3 Scopes and Limitations

Scope:

- Sports events
- School/college tournaments
- Small competitions

Limitations:

- No database storage
- Limited scalability
- Manual data entry required

3.SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before the actual implementation. In this phase, the system is divided into different modules, and the workflow of the application is defined. Proper system design helps in organizing the program efficiently and makes development easier.

The **Sports Tournament Team Registration and Match Result System** is designed as a simple graphical user interface (GUI)-based application using JavaFX, where users can register teams and record match results effectively.

The system is divided into several modules to perform specific tasks. Each module is responsible for handling a particular function of the system, such as managing team registration, recording match results, and displaying tournament details.

3.1 Modules

The main modules of the **Sports Tournament Team Registration and Match Result System** are explained below:

1. Team Registration Module

This module is responsible for storing and managing team details such as team name.

It helps the user maintain a proper list of all participating teams in the tournament

2. **Match Management Module**

This module allows the user to create and manage matches between teams.

3. Result Recording Module

This module is used to record and display match result .

4. **Display Module**

This module displays:
• List of registered teams
• List of match results

5. Validation Module

This module ensures that:
• Input fields are not empty
• Valid data is entered

6.**Exit Module**

This module allows the user to exit the application safely after completing all operations.

4.METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the **Sports Tournament Team Registration and Match Result System**, a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

• Problem Analysis
• System Design
• Program Development
• Testing and Validation
• Result and Documentation

4.1. Problem Analysis

In this phase, the problem is studied and analyzed in detail. The requirements of the system are identified based on the needs of managing sports tournaments, team registrations, and match results.

The main problems identified in manual systems are:

• Difficulty in managing team registrations
•Confusion in recording match results
• Lack of proper data organization
• Errors in maintaining tournament records

Based on these problems, the system requirements were defined to develop an automated application that simplifies tournament management.

4.2. System Design

In this stage, the overall structure of the system is designed before coding begins.

The system is divided into modules such as:

• Team Registration Module
• Match Management Module
• Result Recording Module
• Display Module
•Validation Module
• Exit Module

The program flow and structure are designed so that users can easily interact with the system.

4.3. Program Development

After designing the system, the program is implemented using the Java programming language with JavaFX.

The following steps were followed during development:

• Creation of classes and methods
• Implementation of GUI components
• Writing code for team registration and match result
• Handling user input
• Displaying team and match details

Core Java concepts such as classes, objects, loops, conditional statements, and event handling were used during development.

4.4. Testing and Validation

Testing ensures that the program works correctly and produces the expected results.

Different test cases were used to check the functionality of the system.

Example Test Cases:

Test Case	**Expected Result**
Add Team	Team added successfully
Add Match result	Match stored correctly
Empty input	Error handled properly
Invalid Score	Warning message shown

This phase ensures that the program runs without errors and produces accurate results.

4.5. Result and Documentation

After successful testing, the results of the system are documented. The system manages team registrations and match results effectively.

The results are demonstrated through program execution and screenshots showing input and output.

Proper documentation is prepared to explain the design, implementation, and working of the system.\

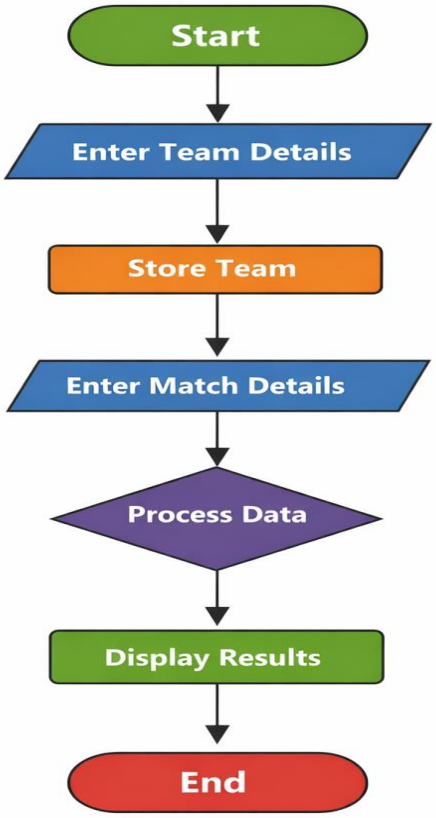
4.6. Algorithm

Algorithm for the Sports Tournament System:

1. Start the program
2. Enter team name
3. Add team to the list
4. Enter match details (Team 1, Team 2, Score)
5. Store match result

6. Display teams and results
7. Generate report if required
8. Repeat if required
9. End the program

4.7. Flowchart



5.SOURCE CODE

Source code:

```
package com.mycompany.sportsmen;
import javafx.application.Application;
import javafx.geometry.Insets;
import javafx.scene.Scene;
import javafx.scene.control.*;
import javafx.scene.layout.*;
import javafx.stage.Stage;
import java.util.*;
import java.util.ArrayList;
import java.util.HashMap;
import java.util.List;
import java.util.Map;
import javafx.application.Application;
import static javafx.application.Application.launch;
import javafx.geometry.Insets;
import javafx.scene.Scene;
import javafx.scene.control.Alert;
import javafx.scene.control.Button;
import javafx.scene.control.ComboBox;
import javafx.scene.control.Label;
import javafx.scene.control.ListView;
import javafx.scene.control.TextField;
import javafx.scene.layout.HBox;
import javafx.scene.layout.VBox;
```

```
import javafx.stage.Stage;
import java.util.ArrayList;
import java.util.HashMap;
import java.util.List;
import java.util.Map;
import javafx.application.Application;
import static javafx.application.Application.launch;
import javafx.geometry.Insets;
import javafx.scene.Scene;
import javafx.scene.control.Alert;
import javafx.scene.control.Button;
import javafx.scene.control.ComboBox;
import javafx.scene.control.Label;
import javafx.scene.control.ListView;
import javafx.scene.control.TextField;
import javafx.scene.layout.HBox;
import javafx.scene.layout.VBox;
import javafx.stage.Stage;
public class App extends Application {
    ListView<String> teamList = new ListView<>();
    ListView<String> matchList = new ListView<>();
    ListView<String> playerList = new ListView<>();
    Map<String, List<String>> teamMembers = new HashMap<>();
    @Override
    public void start(Stage stage) {
        Label title = new Label("Sports Tournament System");
        title.setStyle("-fx-font-size: 18px; -fx-font-weight: bold;");
```

```
TextField teamField = new TextField();
teamField.setPromptText("Enter Team Name");
Button addTeamBtn = new Button("Add Team");
addTeamBtn.setOnAction(e -> {
    String team = teamField.getText();

    if (!team.isEmpty()) {
        teamList.getItems().add(team);
        teamMembers.put(team, new ArrayList<>());
        teamField.clear();
    }
});
TextField playerField = new TextField();
playerField.setPromptText("Enter Player Name");
Button addPlayerBtn = new Button("Add Player");
addPlayerBtn.setOnAction(e -> {
    String selectedTeam = teamList.getSelectionModel().getSelectedItem();
    String player = playerField.getText();

    if (selectedTeam != null && !player.isEmpty()) {
        teamMembers.get(selectedTeam).add(player);
        playerField.clear();
        updatePlayerList(selectedTeam);
    } else {
        showAlert("Select a team and enter player name!");
    }
});
```

```
teamList.setOnMouseClicked(e -> {
    String selectedTeam = teamList.getSelectionModel().getSelectedItem();
    if (selectedTeam != null) {
        updatePlayerList(selectedTeam);
    }
});
VBox teamBox = new VBox(10,
        new Label("Register Team"),
        teamField,
        addTeamBtn,
        teamList,
        new Label("Add Players"),
        playerField,
        addPlayerBtn,
        new Label("Team Members"),
        playerList
);
teamBox.setPadding(new Insets(10));
teamBox.setStyle("-fx-border-color: black;");
ComboBox<String> gameBox = new ComboBox<>();
gameBox.getItems().addAll("Football", "Cricket", "Kabaddi");
gameBox.setPromptText("Select Game");
TextField team1Field = new TextField();
team1Field.setPromptText("Team 1");
TextField team2Field = new TextField();
team2Field.setPromptText("Team 2");
TextField score1Field = new TextField();
```

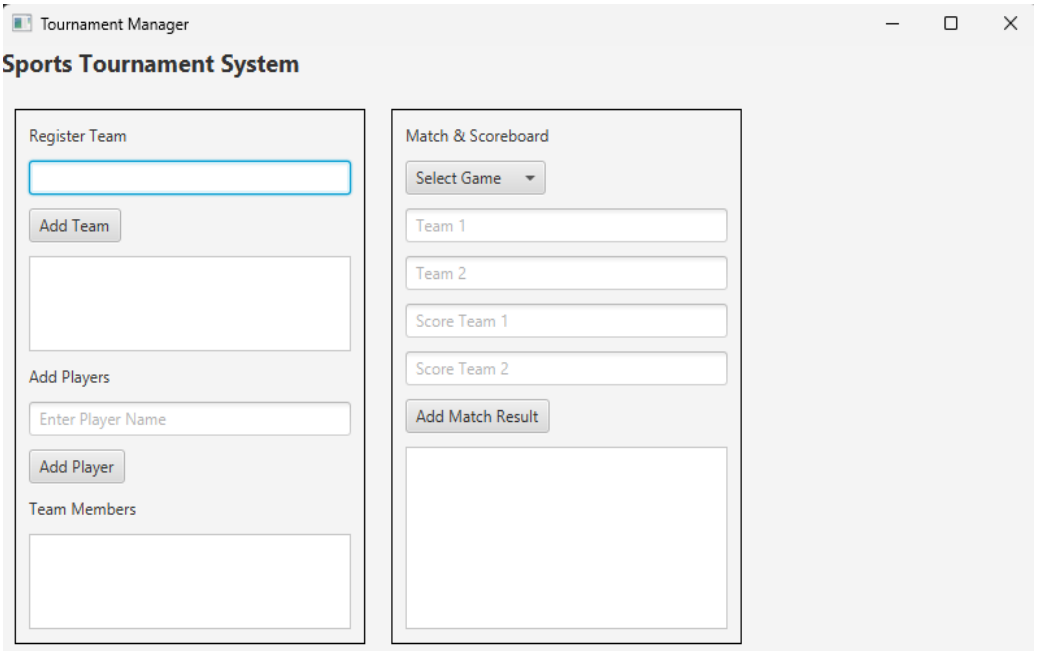
```
score1Field.setPromptText("Score Team 1");
TextField score2Field = new TextField();
score2Field.setPromptText("Score Team 2");
Button addMatchBtn = new Button("Add Match Result");
addMatchBtn.setOnAction(e -> {
   try {
      String game = gameBox.getValue();
      String t1 = team1Field.getText();
      String t2 = team2Field.getText();
      int s1 = Integer.parseInt(score1Field.getText());
      int s2 = Integer.parseInt(score2Field.getText());
      String winner;
      if (s1 > s2) {
         winner = t1;
      } else if (s2 > s1) {
         winner = t2;
      } else {
         winner = "Draw";
      }
      matchList.getItems().add(
            game + ": " + t1 + " (" + s1 + ") vs "
                  + t2 + " (" + s2 + ") → Winner: " + winner
      );
      team1Field.clear();
      team2Field.clear();
      score1Field.clear();
      score2Field.clear();
```

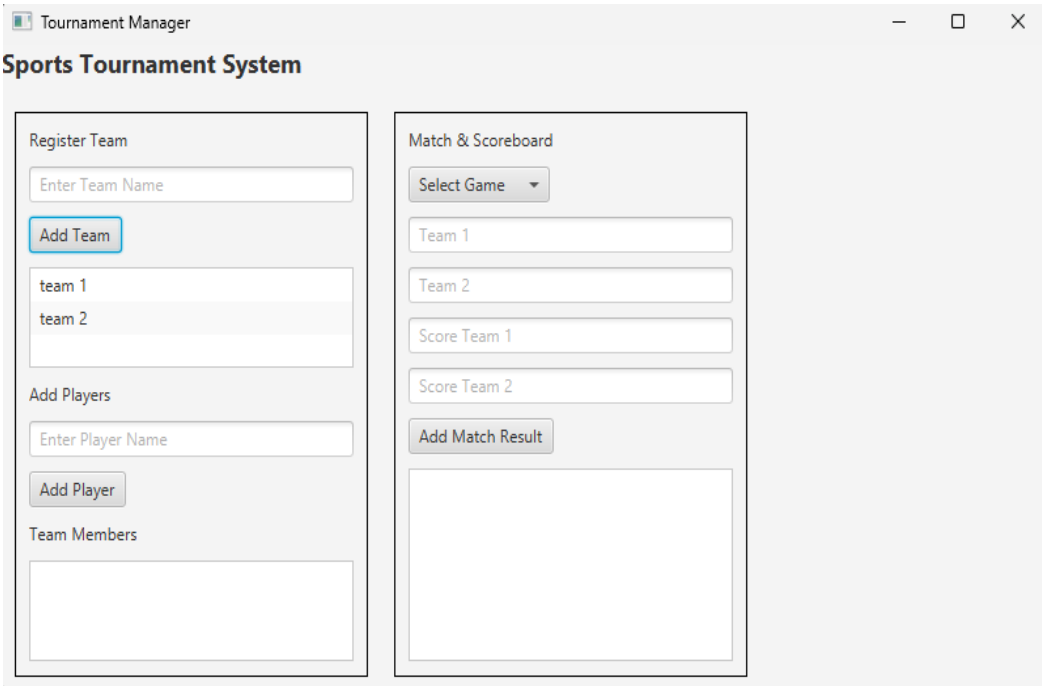
```
        } catch (Exception ex) {
            showAlert("Enter valid scores!");
        }
    });
    VBox matchBox = new VBox(10,
            new Label("Match & Scoreboard"),
            gameBox,
            team1Field,
            team2Field,
            score1Field,
            score2Field,
            addMatchBtn,
            matchList
    );
    matchBox.setPadding(new Insets(10));
    matchBox.setStyle("-fx-border-color: black;");
    HBox mainLayout = new HBox(20, teamBox, matchBox);
    mainLayout.setPadding(new Insets(10));
    VBox root = new VBox(10, title, mainLayout);
    Scene scene = new Scene(root, 800, 450);
    stage.setScene(scene);
    stage.setTitle("Tournament Manager");
    stage.show();
}
private void updatePlayerList(String team) {
    playerList.getItems().clear();
    playerList.getItems().addAll(teamMembers.get(team));
```

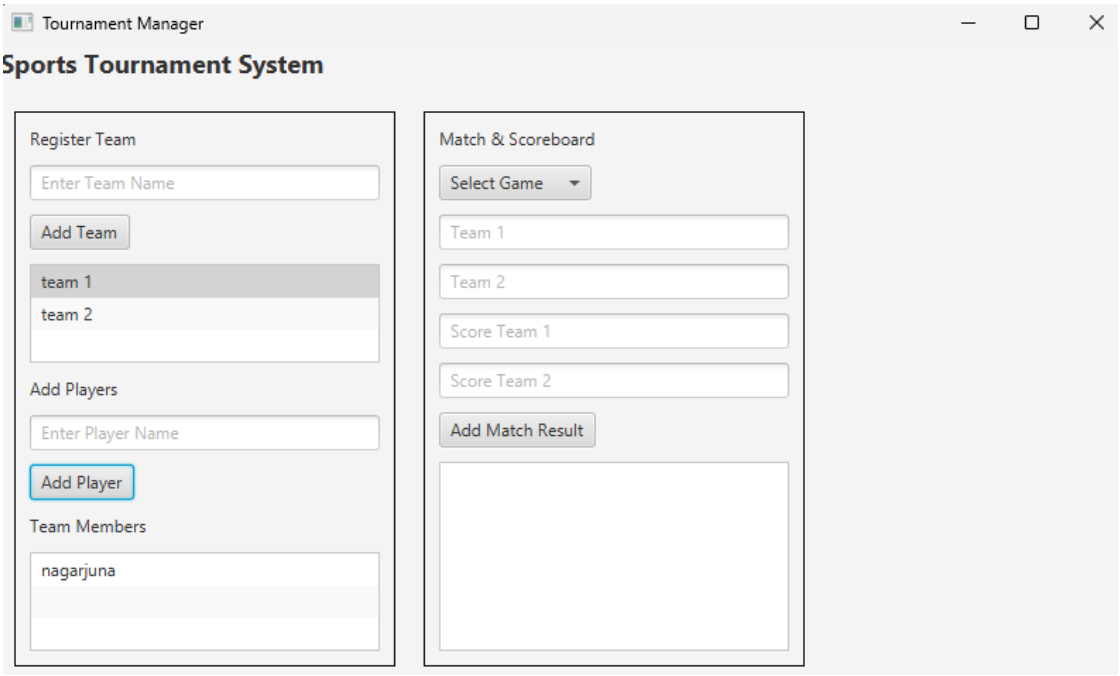
```
    }
    private void showAlert(String msg) {
        Alert alert = new Alert(Alert.AlertType.ERROR);
        alert.setContentText(msg);
        alert.show();
    }
    public static void main(String[] args) {
        launch();
    }
}
```

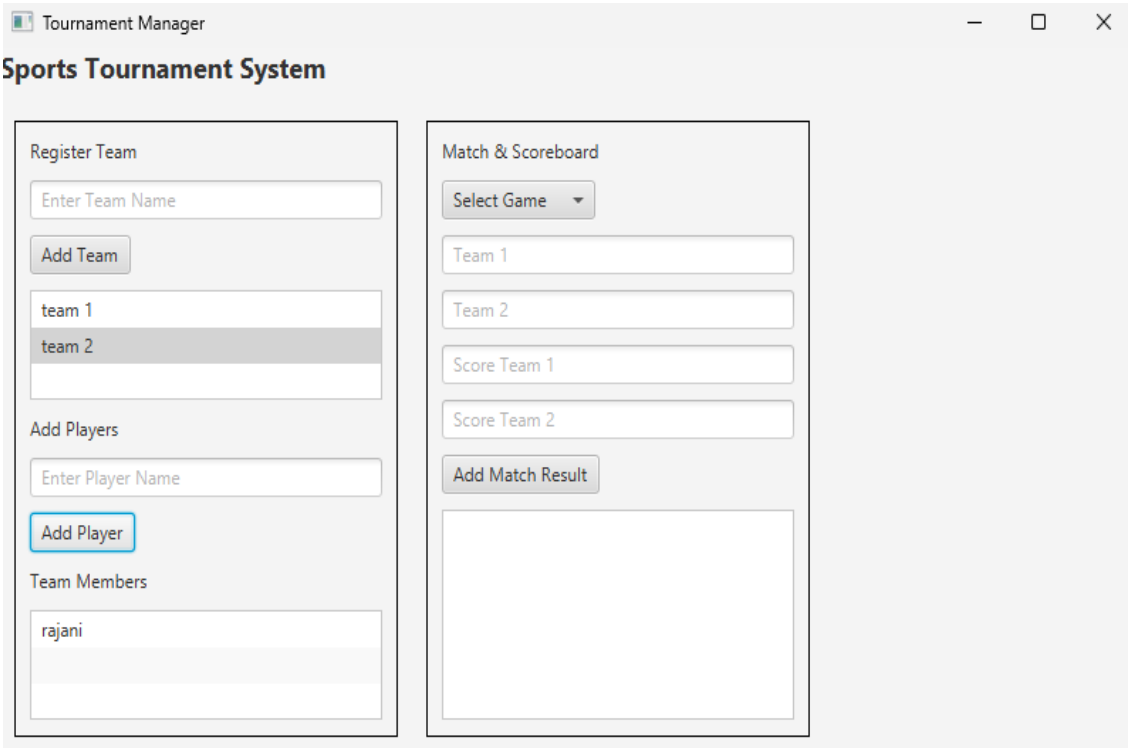
6. INPUT & OUTPUT SCREENS

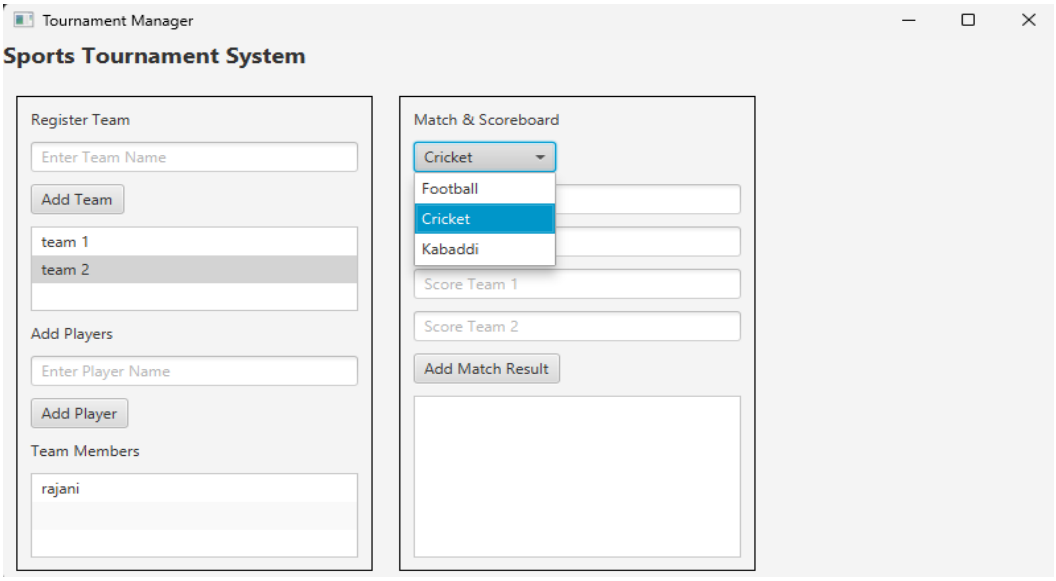
Input Screen :

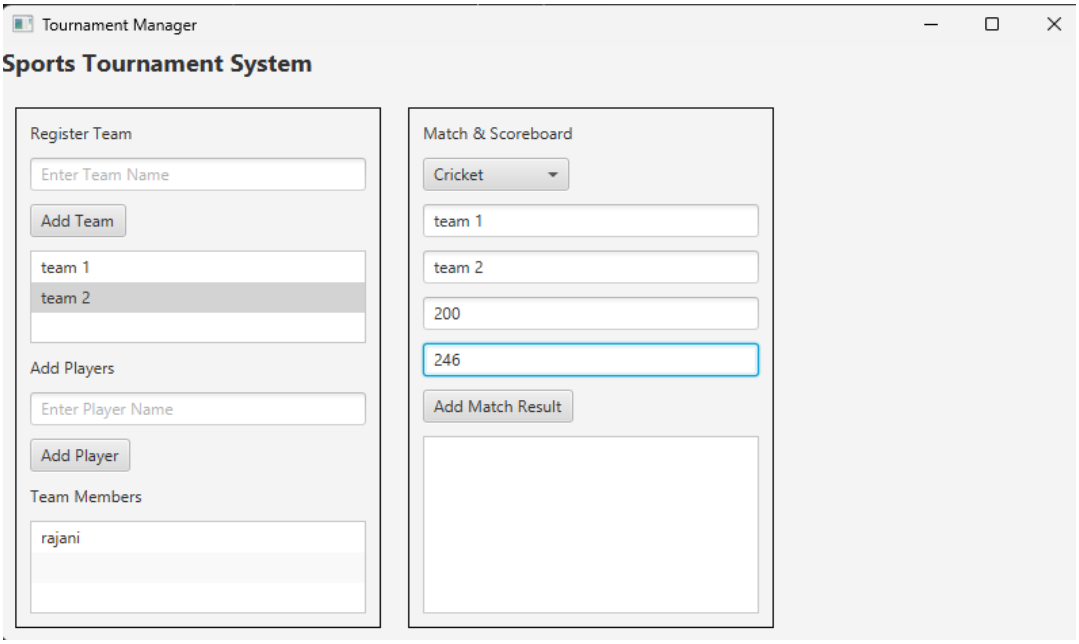




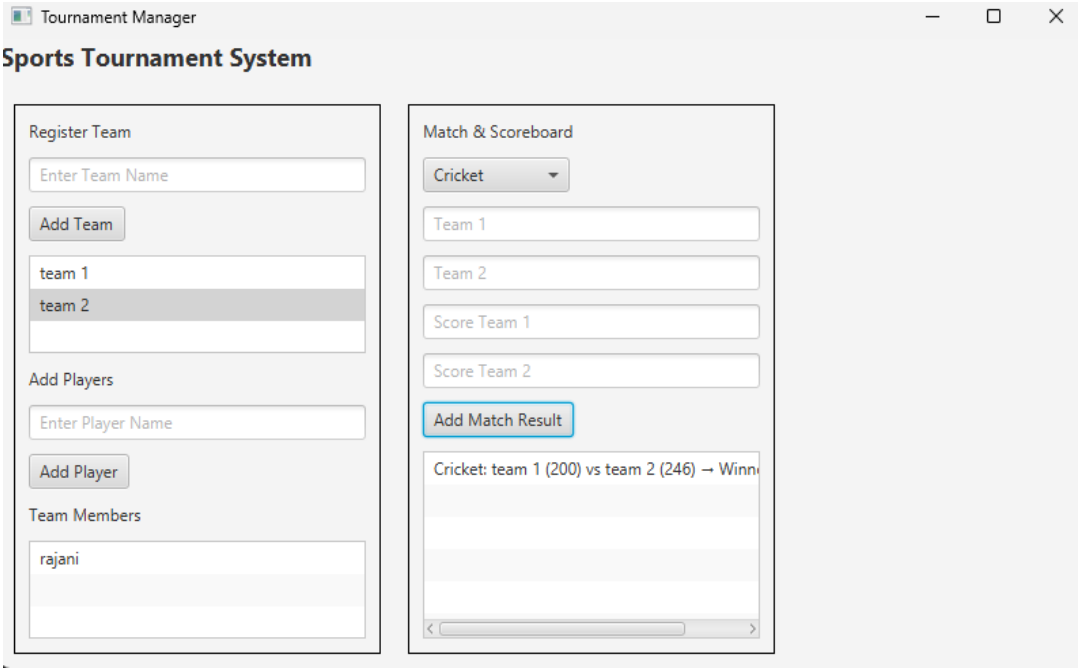

Tournament Manager
Sports Tournament System
Register Team
Enter Team Name
Add Team
team 1
team 2
Add Players
Enter Player Name
Add Player
Team Members
nagarjuna
Match & Scoreboard
Select Game
Team 1
Team 2
Score Team 1
Score Team 2
Add Match Result


Tournament Manager
Sports Tournament System
Register Team
Enter Team Name
Add Team
team 1
team 2
Add Players
Enter Player Name
Add Player
Team Members
rajani
Match & Scoreboard
Select Game
Team 1
Team 2
Score Team 1
Score Team 2
Add Match Result


Tournament Manager
Sports Tournament System
Register Team
Enter Team Name
Add Team
team 1
team 2
Add Players
Enter Player Name
Add Player
Team Members
rajani
Match & Scoreboard
Cricket
Football
Cricket
Kabaddi
Score Team 1
Score Team 2
Add Match Result


Tournament Manager
Sports Tournament System
Register Team
Enter Team Name
Add Team
team 1
team 2
Add Players
Enter Player Name
Add Player
Team Members
rajani
Match & Scoreboard
Cricket
team 1
team 2
200
246
Add Match Result

Output Screen :



7.DISCUSSION & CONCLUSION

The **Sports Tournament Team Registration and Match Result System** was developed to simplify the process of managing sports tournaments by handling team registrations and recording match results efficiently. The system allows users to add team details, store match results, and display tournament information in an organized manner.

During the development process, Core Java programming concepts such as classes, objects, loops, conditional statements, and event handling were applied. The system successfully manages team data and accurately records match results.

The system was tested with different inputs to ensure that it performs all operations correctly. The results show that the system reduces manual effort, improves organization, and helps in managing tournaments more efficiently.

7.2 Conclusion

The **Sports Tournament Team Registration and Match Result System** successfully demonstrates the practical application of JavaFX and Core Java concepts in developing a real-world application.

The system simplifies team registration and match result management, reduces errors, and improves efficiency. It provides a structured way to organize tournament data.

This project helps students understand how Java programming can be used to develop practical software systems. The project can be further enhanced by adding features such as database connectivity, advanced user interface design, cloud storage, and online tournament tracking.

8.REFERENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – *Java: The Complete Reference*

Websites

• https://www.oracle.com/java
• https://netbeans.apache.org/download/index.html